



ORGANIC CHEMISTRY

Carbon compounds • Homologous series • Reactions • Polymers • Exam essentials

PrepAce

One-Page Revision Cheatsheet

1 The Basics

- Organic compounds** contain **carbon**, bonded mainly to H, O, N, halogens. C forms **4 covalent bonds** (chains, branches, rings).
- Hydrocarbon**: only C and H. **Saturated** = only C–C single bonds. **Unsaturated** = at least one C=C or C≡C.
- Homologous series**: same functional group and general formula; each member differs by CH₂; similar chemistry; boiling point rises with chain length.
- Functional group**: the atom/group that decides the chemical properties.

1	2	3	4	5	6	7	8	9	10
meth	eth	prop	but	pent	hex	hept	oct	non	dec

2 Homologous Series at a Glance

Series	Formula	Group	Ending	Example
Alkane	C _n H _{2n+2}	C–C	-ane	C ₂ H ₆ ethane
Alkene	C _n H _{2n}	C=C	-ene	C ₂ H ₄ ethene
Alkyne	C _n H _{2n-2}	C≡C	-yne	C ₂ H ₂ ethyne
Haloalkane	C _n H _{2n+1} X	–Cl, –Br	halo-	CH ₃ Cl chloromethane
Alcohol	C _n H _{2n+1} OH	–OH	-ol	C ₂ H ₅ OH ethanol
Carboxylic acid	RCOOH	–COOH	-oic acid	CH ₃ COOH ethanoic acid
Ester	RCOOR'	–COO–	-yl ... -oate	CH ₃ COOC ₂ H ₅ ethyl ethanoate

3 Naming & Isomers

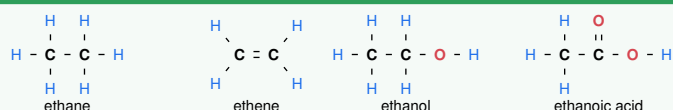
- Find the **longest carbon chain**: this gives the stem (meth, eth, prop...).
- Add the **ending** for the functional group (-ane, -ene, -ol, -oic acid).
- Number** the chain so the functional group (or C=C) gets the **lowest number**.
- Name branches as prefixes (methyl-, ethyl-) in alphabetical order; use di-, tri- for repeats.

Examples: CH₂=CHCH₂CH₃ = **but-1-ene**

CH₃CH(CH₃)CH₂CH₃ = **2-methylbutane**

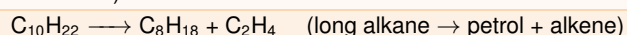
Isomers: same molecular formula, **different structure**. E.g. C₄H₁₀: **butane** and **2-methylpropane**.

4 Displayed Formulae

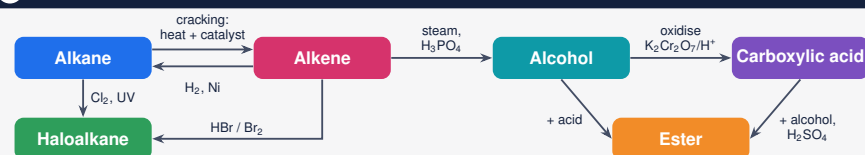


5 Crude Oil: Distillation & Cracking

- Fractional distillation** separates crude oil by **boiling point**. Column is hot at the bottom, cool at the top.
- Fractions (top → bottom)**: refinery gases → petrol → naphtha → kerosene → diesel → fuel oil → bitumen.
- Going down**: longer chains, **higher** boiling point, **more viscous**, **less flammable**, smokier flame.
- Cracking**: breaks long alkanes into shorter, more useful alkanes + alkenes. Needs **high temperature (500–600 °C)** + **catalyst** (silica/alumina) or steam.



* Reaction Map: How Compounds Link Together



6 Alkanes (Saturated, Fairly Unreactive)

- Complete combustion**: CH₄ + 2 O₂ → CO₂ + 2 H₂O
- Incomplete combustion** (limited O₂) gives CO (toxic) and/or C (soot) + H₂O
- Substitution** (UV light): CH₄ + Cl₂ $\xrightarrow{\text{UV}}$ CH₃Cl + HCl

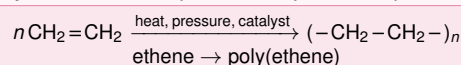
7 Alkenes (Unsaturated, Reactive C=C)

Addition reactions: the double bond opens and two atoms add on.

Reagent	Conditions	Product
H ₂	Ni catalyst, 150 °C	alkane (hydrogenation)
Br ₂ or Cl ₂	room temperature	dihaloalkane
HBr or HCl	room temperature	haloalkane
H ₂ O (steam)	H ₃ PO ₄ , 300 °C, 60 atm	alcohol (hydration)

Test for alkenes: shake with **bromine water**: **orange** → **colourless**. Alkanes: stays orange.

Addition polymerisation: many monomers join, C=C opens, nothing is lost.



Propene → poly(propene); chloroethene → PVC. **Condensation polymers** (nylon, polyesters) use 2 monomers and lose H₂O. Most plastics are **not biodegradable**.

8 Alcohols (–OH) and Ethanol

	Fermentation	Hydration of ethene
Reaction	C ₆ H ₁₂ O ₆ $\xrightarrow{\text{yeast}}$ 2 C ₂ H ₅ OH + 2 CO ₂	C ₂ H ₄ + H ₂ O $\rightleftharpoons$ C ₂ H ₅ OH
Conditions	25–35 °C, no oxygen	300 °C, 60 atm, H ₃ PO ₄
Source	sugar (renewable)	crude oil (finite)
Rate/purity	slow, batch, impure	fast, continuous, pure

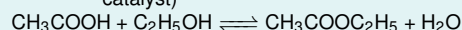
- Combustion**: C₂H₅OH + 3 O₂ → 2 CO₂ + 3 H₂O (clean fuel)
- Oxidation** (acidified K₂Cr₂O₇ or KMnO₄) → **carboxylic acid**; dichromate goes **orange** → **green**.
- Reacts with Na to give H₂ gas (fizzing). Uses: fuel, solvent, drinks.

9 Carboxylic Acids & Esters

- Weak acids** (partly ionise), pH about 3–5. Typical acid reactions:

With	Gives
Metal (Mg)	salt + H ₂ : Mg + 2 CH ₃ COOH → (CH ₃ COO) ₂ Mg + H ₂
Carbonate	salt + H ₂ O + CO ₂ (fizzing; limewater turns milky)
Base/alkali	salt + water (ethanoates)

Esterification: acid + alcohol $\rightleftharpoons$ ester + water (warm, conc. H₂SO₄ catalyst)



Esters smell **sweet and fruity**: perfumes and flavourings. Name: alcohol part first (ethyl), acid part second (ethanoate).

10 Quick Tests

Test for	Observation
Alkene (C=C)	bromine water: orange → colourless
Alcohol	acidified K ₂ Cr ₂ O ₇ : orange → green (on warming)
Carboxylic acid	Na ₂ CO ₃ : fizzes, CO ₂ turns limewater milky
Complete combustion	clean blue flame, CO ₂ + H ₂ O only

! Exam Tips & Worked Example

- Balance**: C₃H₈ + 5 O₂ → 3 CO₂ + 4 H₂O
- Formula**: alkene with 6 C → C_nH_{2n}: C₆H₁₂
- Gas volumes (r.t.p.)**: 1 mol = 24 dm³; gases react in mole ratio, so 10 cm³ CH₄ needs 20 cm³ O₂.
- Always quote **conditions** (catalyst, T, P).
- Say **“colourless”**, not “clear”, in bromine tests.